A MINIMAL NON-SCHEME CLOSED SUBSPACE EXISTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4027-4042, following the statement at 4019-4025.

Editorial note. No omissions; the empty-case remark is reproduced in the counted notes.

Allowed standard prerequisites (editorial).

- Limits that are schemes. lemma-limit-is-scheme (same chapter, cited at ~4035-4036). If Z = lim Z_alpha were a scheme, one of the Z_alpha would be a scheme.
- Limits are nonempty. lemma-limit-nonempty (same chapter, cited at ~4037-4039). Rules out Z = emptyset in the chain argument.

Source reference. Original lemma 07SR: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1435–1435. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 086W: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1189–1189. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. If X is not a scheme, then there exists a closed subspace $Z \subset X$ such that Z is not a scheme, but every proper closed subspace $Z' \subset Z$ is a scheme.*

Proof. We prove this by Zorn's lemma. Let $\mathcal{Z}$ be the set of closed subspaces Z which are not schemes ordered by inclusion. By assumption $\mathcal{Z}$ contains X, hence is nonempty. If Z_α is a totally ordered subset of $\mathcal{Z}$, then $Z = \bigcap Z_\alpha$ is in $\mathcal{Z}$. Namely,

$$Z = \lim Z_\alpha$$

and the transition morphisms are affine. Thus we may apply Lemma 07SR to see that if Z were a scheme, then so would one of the Z_α. (This works even if $Z = \emptyset$, but note that by Lemma 086W this cannot happen.) Thus $\mathcal{Z}$ has minimal elements by Zorn's lemma. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.